

The Hong Kong Polytechnic University

Subject Description Form

Please read the notes at the end of the table carefully before completing the form.

Subject Code	SFT547
Subject Title	Smart Fashion and Textiles
Credit Value	3
Level	5
Pre-requisite/ requisite/ Exclusion	N/A
Objectives	This subject aims at providing students with knowledge in the field of smart fashion and textiles through case studies and analysis, from market requirement to design process, wear performance and potential emerging applications. It gives an insightful overview of smart fashion and textile technologies through the introduction of a wide range of cutting-edge projects in the fields.
Intended Learning Outcomes <i>(Note 1)</i>	Upon completion of the subject, students will be able to: a) have the knowledge and information about the latest smart fibers, fabrics, and garments b) demonstrate their holistic knowledge and understanding of the main functional properties and potential applications of smart fashion and textiles c) develop relevant criteria and technical specifications for the new smart products which will satisfy the requirements of the target market and wear performance without compromising their original functionality d) perform analysis of the impact of the technological development of smart fashion and textiles on the industry and market
Subject Synopsis/ Indicative Syllabus <i>(Note 2)</i>	1. User-centered product development This topic introduces the basic development process of smart fashion and textiles, including conducting market research, defining the problem, proposing design strategies, generating ideas, developing and designing smart products, and evaluating function and wear performance.

2. Introduction to smart materials for fashion and textiles

This topic briefly introduces different types of smart materials, including conductive fibres and fabrics, thermo-, humidity-, electric-, electromagnetic-, pressure-, pH-, and gas-sensitive fabrics and clothing, biomimetic fibres and fabrics, phase change materials, wireless and wired fabric sensors and actuators, etc.

3. Sustainable manufacture of fashion and textiles with intelligent properties

This topic discusses knitting, weaving, moulding, and 3D printing methods of fabricating smart fashion and textiles, using sustainable functional materials and integrating wearable devices and electronics.

4. Performance evaluation

This guided study discusses the fundamentals and practices of the evaluation of smart textile and clothing products, especially on the consideration of wear comfort in addition to their interactive functions.

The performance requirements include physical comfort, physiological and psychological comfort, handle property, making-up quality, wear performance (e.g., dimensional stability, appearance retention, durability and aftercare), etc.

5. Case analysis: thermoregulation

This topic covers different types of smart fashion and textiles for thermal and moisture management, including active modes (e.g., fan air-conditioned clothing, thermoelectric fabric, circulating liquid garment, and electric heating clothing) and passive modes (e.g., evaporative cooling vest, radiation cooling fabric, and sweat-stimuli adaptive fabric). The subjective and objective evaluation methods will be discussed.

6. Case analysis: chemical/biological protection

This topic introduces smart chemical/biological protective clothing made from high-efficiency but breathable filter fabrics and incorporated with catalysts and antimicrobial treatments. The evaluation methods will be introduced.

7. Case analysis: sensor and actuator

	This topic covers different types of smart fashion and textiles that function as sensors or boost connectivity between wearable sensors (e.g., stress sensor, strain sensor, humidity sensor, temperature sensor, sound sensor, and heart rate sensor), and integrate flexible actuators on the basis of the latest technologies such as RFID and NFC wireless communication, electrochemical actuation, 3D printing, and shape memory materials. 8. Case analysis: human-environment interface This topic discusses the latest development of smart clothing as human-environment interface for connecting human with clouds and big data for sustainable health monitoring, social communication, navigation, and visible screen. 9. Case analysis: augmented human-body functions This topic discusses the latest development in smart fashion and textiles for enhancing human-body functions, such as powered exoskeleton armour, shape-memory-alloy-based fabric muscle, sharkskin swimsuit, and multifrequency optical invisibility cloak. 10. Case analysis: aesthetic enhancement This topic discusses the latest development in smart fashion and textiles for aesthetic enhancement, mainly including fabrics that light up and fabrics that can change colour or shape in response to sound or heat. The color changing and lighting scheme can also work by embedding the fabric with wearable electronics. 11. Case analysis: medical care This topic discusses the latest development of smart textiles for medical care, such as breathable and biodegradable wound dressing, drug-based release systems, and heart rhythm diagnostics.
Teaching/Learning Methodology (Note 3)	The subject matter is mainly delivered by lectures. Student-centred tutorials provide more interactive learning. The subject matters will be reinforced by group discussions, small projects, and practical demonstrations.

Assessment Methods in Alignment with Intended Learning Outcomes (Note 4)	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)			
			a	b	c	d
	1. Group Project	30%	✓	✓	✓	
	2. Presentation	20%	✓	✓	✓	
	3. Examination	50%	✓	✓	✓	✓
	Total	100 %				
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:					
	The group projects require students to analyze and address market and consumers’ demand and develop smart fashion and textile products with relevant technical specifications. In-class presentations and discussions will provide students with a better understanding of smart fashion and textile development for various applications. Students’ understating and knowledge in terms of the concept of smart fashion and textiles, materials and technology specifications, market requirements, critical thinking, and analytical skills will be assessed by written examination.					
	The materials submitted for this assessment must be the student’s own work. The submitted work may not be accepted for the purpose of assessment if its authenticity is questionable. Submitting GenAI-generated materials as students’ own work or part of their work is an act of academic dishonesty. Students who are found committing academic dishonesty will face disciplinary actions.					
Student Study Effort Expected	Class contact:					
	▪ Lecture					26Hrs.
	▪ Tutorials					12Hrs.
	Other student study effort:					
	▪ Reading					67 Hrs.
	Total student study effort					105Hrs.

Reading List and References	Reference Book Xiaoming Tao. Handbook of Smart Textiles. Springer, 2015 Schneegass, Stefan; Amft, Oliver. Smart Textiles: Fundamentals, Design, and Interaction. Springer, 2018 Koncar, Vladan; Koncar, Vlada. Smart Textiles and Their Applications. Elsevier, 2016 Jane McCann David Bryson. Smart Clothes and Wearable Technology. Elsevier, 2009 Paret Dominique Pierre Crego. Wearables, Smart Textiles & Smart Apparel. Elsevier, 2018
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Note 1: Intended Learning Outcomes

Intended learning outcomes should state what students should be able to do or attain upon subject completion. Subject outcomes are expected to contribute to the attainment of the overall programme outcomes.

Note 2: Subject Synopsis/Indicative Syllabus

The syllabus should adequately address the intended learning outcomes. At the same time, overcrowding of the syllabus should be avoided.

Note 3: Teaching/Learning Methodology

This section should include a brief description of the teaching and learning methods to be employed to facilitate learning, and a justification of how the methods are aligned with the intended learning outcomes of the subject.

Note 4: Assessment Method

This section should include the assessment method(s) to be used and its relative weighting, and indicate which of the subject intended learning outcomes that each method is intended to assess. It should also provide a brief explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes.